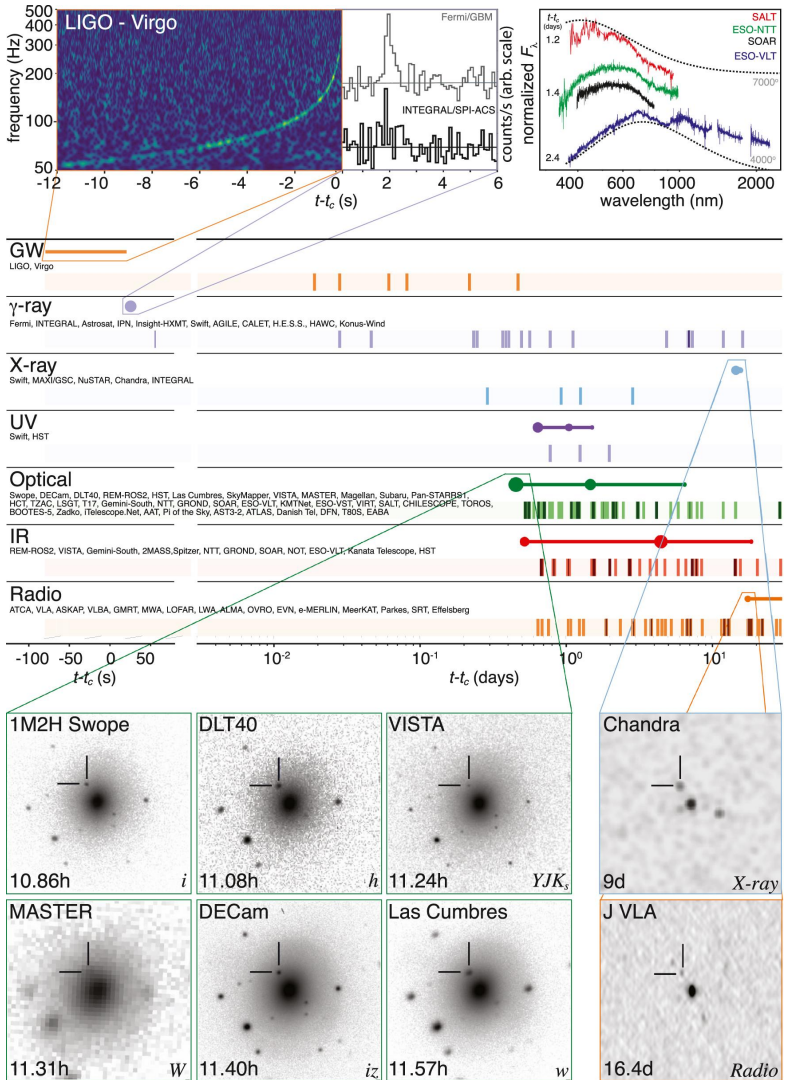
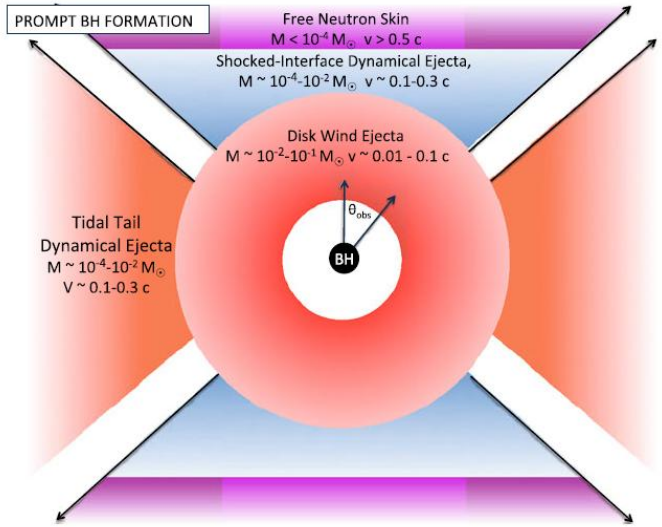
# 使用 SoXS 在 LSST 时代下搜寻引力波电磁对应体

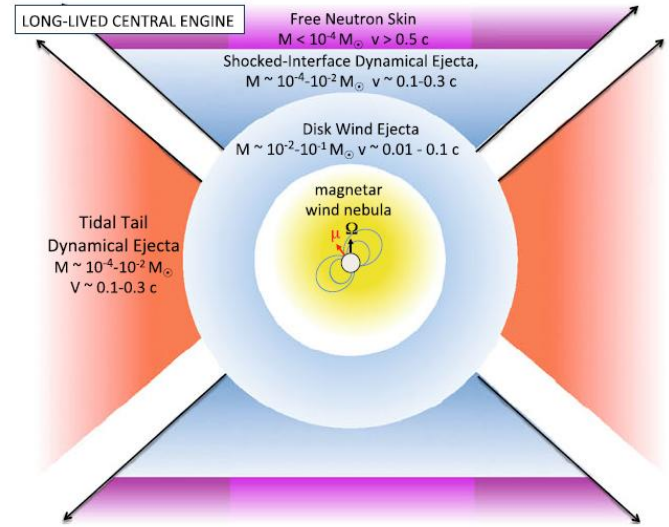
周浩 PMO

2026年9月28日星期一

# 引力波电磁对应体 Kilonova
# GW170817/GRB 170817/AT2017gfo



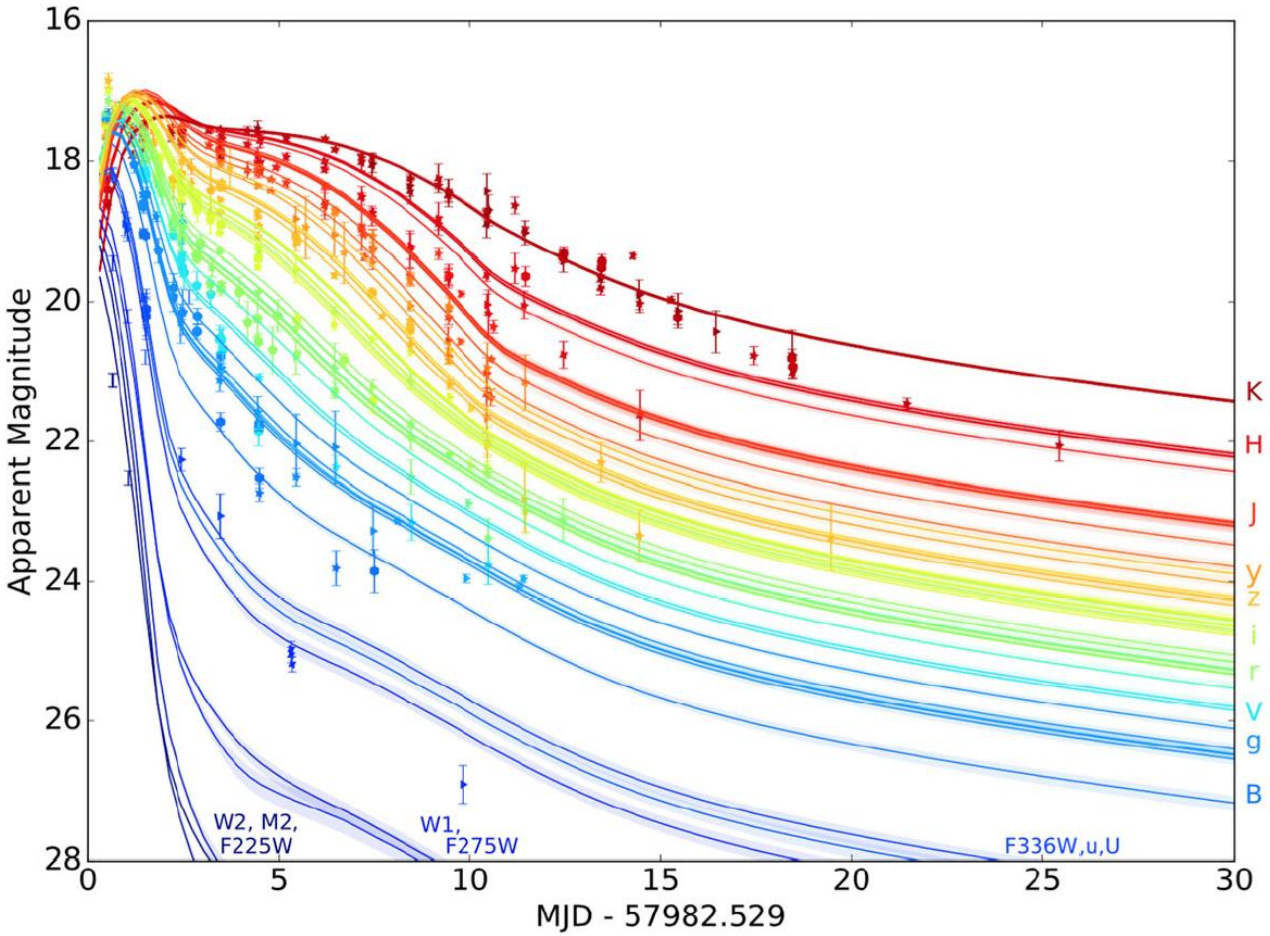



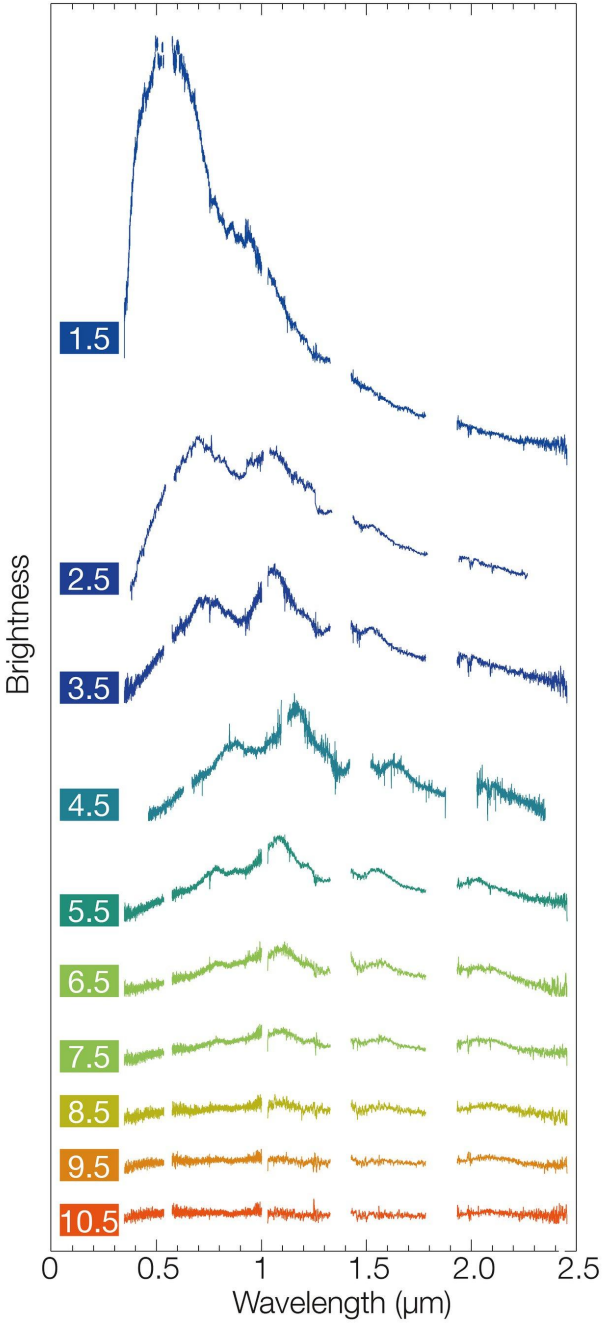

Metzger+ 2019
10.1007/s41114-019-0024-0

Stratta+ 2022
10.3390/universe8090459

# Kilonova



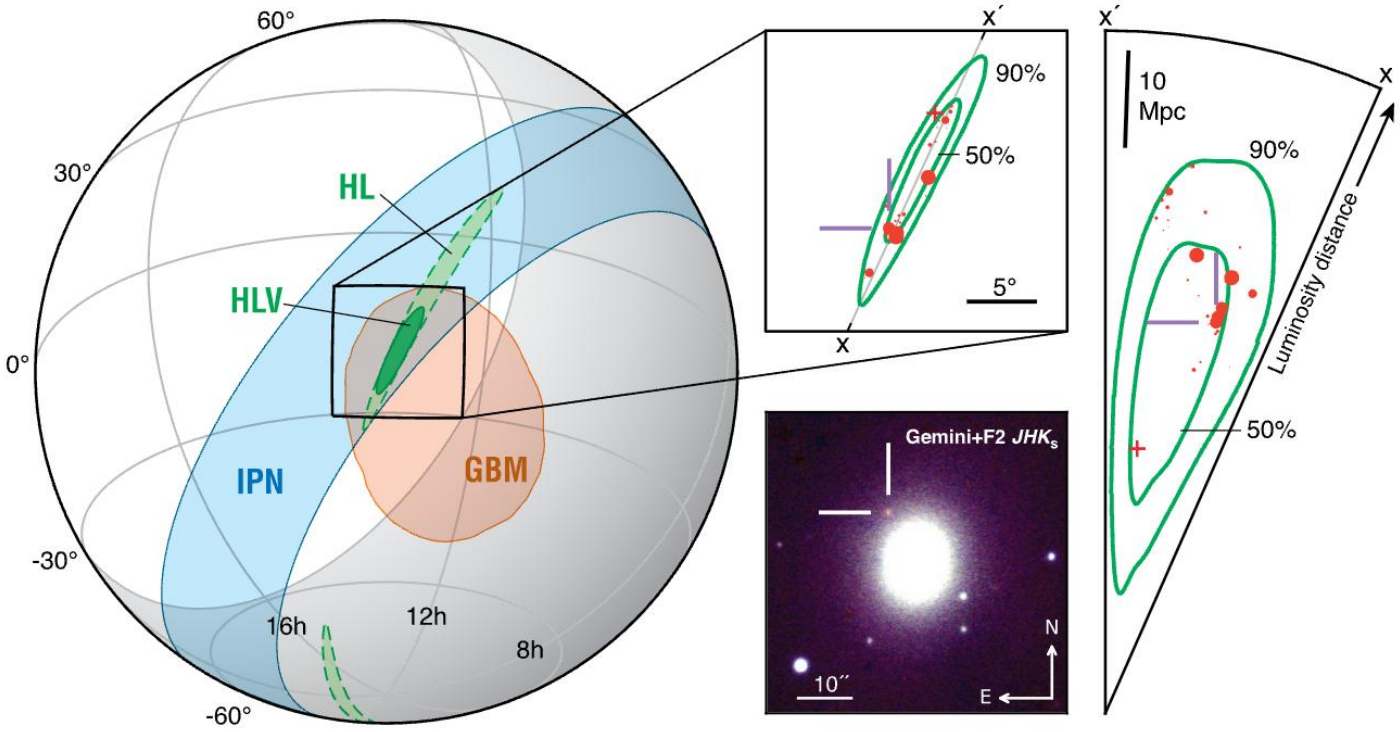

Villar+ 2017
10.3847/2041−8213



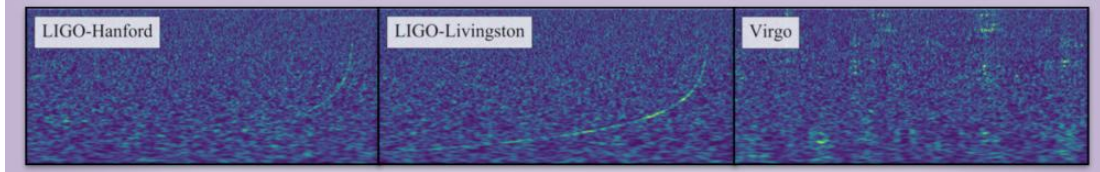

https://www.eso.org/public/images/eso1733j/

# GW alert

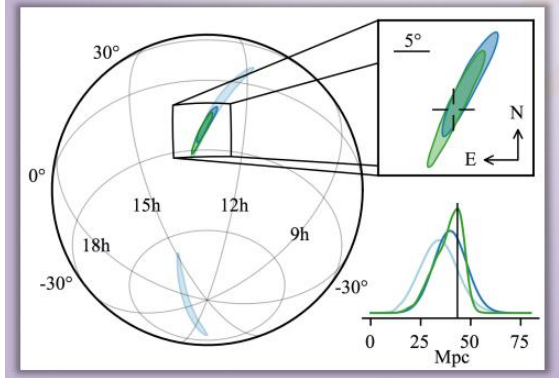
- 巨大的定位误差




## GW170817 FACTSHEET




| | |
|---|---|
| observed by | H, L, V |
| source type | binary neutron star (NS) |
| date | 17 August 2017 |
| time of merger | 12:41:04 UTC |
| signal-to-noise ratio | 32.4 |
| false alarm rate | < 1 in 80 000 years |
| distance | 85 to 160 million light-years |
| total mass | 2.73 to 3.29 $M_\odot$ |
| primary NS mass | 1.36 to 2.26 $M_\odot$ |
| secondary NS mass | 0.86 to 1.36 $M_\odot$ |
| mass ratio | 0.4 to 1.0 |
| radiated GW energy | > 0.025 $M_\odot c^2$ |
| radius of a 1.4 $M_\odot$ NS | likely ≤ 14 km |
| effective spin parameter | -0.01 to 0.17 |
| effective precession spin parameter | unconstrained |
| GW speed deviation from speed of light | < few parts in $10^{15}$ |
| inferred duration from 30 Hz to 2048 Hz** | ~ 60 s |
| inferred # of GW cycles from 30 Hz to 2048 Hz** | ~ 3000 |
| initial astronomer alert latency* | 27 min |
| HLV sky map alert latency* | 5 hrs 14 min |
| HLV sky area† | 28 deg$^2$ |
| # of EM observatories that followed the trigger | ~ 70 |
| also observed in | gamma-ray, X-ray, ultraviolet, optical, infrared, radio |
| host galaxy | NGC 4993 |
| source RA, Dec | 13$^h$09$^m$48$^s$, -23°22'53" |
| sky location | in Hydra constellation |
| viewing angle (without and with host galaxy identification) | ≤ 56° and ≤ 28° |
| Hubble constant inferred from host galaxy identification | 62 to 107 km s$^{-1}$ Mpc$^{-1}$ |




Images: time frequency traces (top), GW sky map (left, HL = light blue, HLV = dark blue, improved HLV = green, optical source location = cross-hair)

GW=gravitational wave, EM = electromagnetic, $M_\odot$=1 solar mass=2x10$^{30}$ kg, H/L=LIGO Hanford/Livingston, V=Virgo

Parameter ranges are 90% credible intervals.
*referenced to the time of merger
**maximum likelihood estimate
†90% credible region

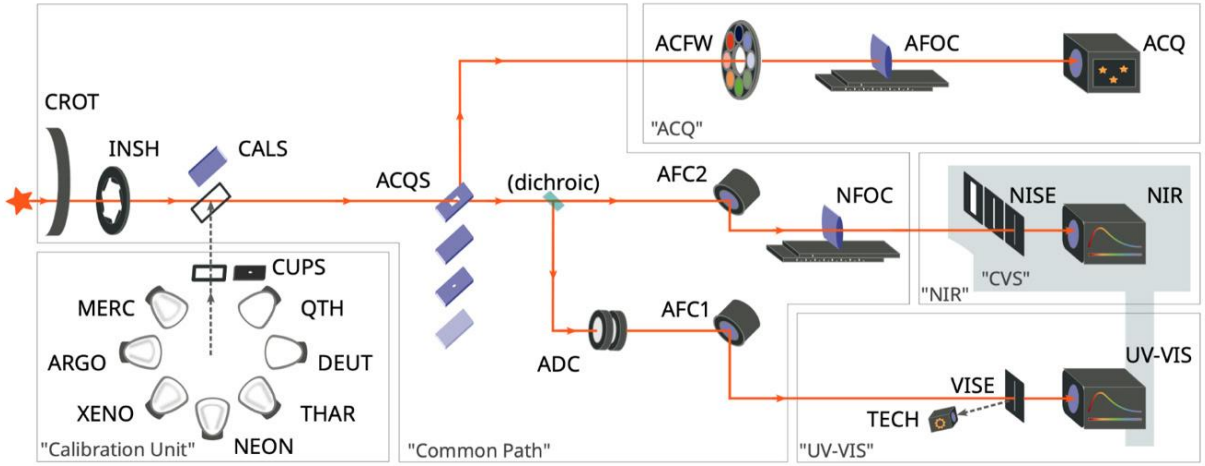
# Son of X-Shooter (SoXS)

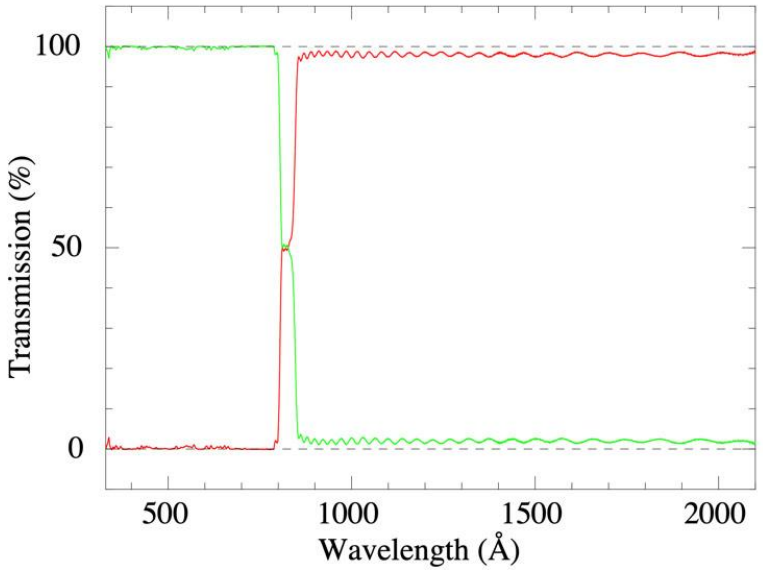
NTT at La Silla



The sun sets behind the New Technology Telescope (NTT) at ESO's La Silla observation site. La Silla, in the southern part of the Atacama desert of Chile, was ESO's first observation site. Home to ESO 3.6-metre telescope and the 3.58-metre New Technology Telescope (NTT), the site sits 2400 metres above sea level, creating excellent viewing conditions for astronomers. La Silla also hosts many national telescopes, including the Swiss 1.2-metre Leonhard Euler Telescope and the Danish 1.54-metre telescope.

**Credit:** Iztok Boncina/ESO




The Acquisition and Guiding system consists of a **3.5'x3.5'** (square) Field of View scientific CCD for target acquisition. A filter wheel equipped with the **ugrizy** SDSS filters is available (together with **Cousin V**).

# SoXS daily log

| 22 Sep 2026 | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Object | RA | Dec | Mag | Spec. | slit w | Filte | Exp (s) | Start (UT | End (UT | Notes Obs |
| | | | | | | | | | | **Observers: Kevin Alabarta and Camila Cárdenas** |
| FLAT | | | | | | | | 23:03:00 | 23:23:00 | Started with sun at -5, ended with sun at -12, the meteor control is freezing and lose the mouse's control. NIR (NISE) is simulate mood, because the cooling was not working. |
| **Mouse doesn't work** | | | | | | | | | | |
| ONECAL | | | | | | | | | | T = 14.3, Alt = 78, M2 = -3.24 |
| IMAGE ANALYSIS | | | | | | | | 23:30:00 | 23:50:00 | We have clouds+moon and the operator used other technique for the image analysis. T = 12.5, Alt = 76, M2 = -3.53 |
| EG274 | 16:23:34 | -39:13:46 | 11.03 | ST | acq | V | 180 | 0:06:00 | 1:05:00 | M2= 5.2 , T=14. Restart the observation for coma effect (Focus problem). New focus with onecal M2= 3.32. The NIR and VIS exp is not the same was for default. |
| IMAGE ANALYSIS | | | | | | | | 0:32:00 | 0:40:00 | We have clouds+moon and the operator used other tecnic for the image analysis close to the zenit. T = 14.03, Alt = 46, M2 = -3.32. The problem was the Alt for the first image analysis with clouds is better to use the zenit. CORRECTION: In the cookbook "If the weather is bad, the operator will do only the ONECAL (no target needed). After this, you must correct the M2 value by asking the operator to add 0.2 mm to the actual M2 value. If we want to check if our VIS focus is consistent with the IMAGE ANALYSIS one, we can run our focus-VIS-template (a). For the acquisition focus, the values for each filter are already implemented in the software (every time you do imaging, the software automatically moves the "afoc" in the correct position according to the filter). If you want to check if they are correct, you can run the focus-IMG-template" |
| AT2026aaam | 18:24:36 | -14:51:13 | 17 | | acq | | 40 | 1:10:00 | 1:25:00 | T = 14.03, M2 = -3.32. We change the exp because the ob is faint. |
| AT2026aaam | 18:24:36 | -14:51:13 | 17 | | 0.5 | | 300 | 1:26:00 | 1:48:00 | T = 14.03, M2 = -3.32. |
| V6125Sct | 18:31:46 | -14:18:55 | 19.2 | SN | img | ugriz | 120 | 1:50:00 | 2:47:00 | T = 13.83, M2 = -3.32. We have bad conditions for the clouds+moon |
| **Close Telescope (2:47:00) :(** | | | | | | | | | | |
| **Open Telescope (3:44:00) :)** | | | | | | | | | | |
| AT2026acgs | 19:56:12 | 8:25:27 | 16.14 | SN | 1 | | 300 | 3:48:00 | 4:16:35 | Stare mode, T=13.8, M2= -3.32. Have a difference in exp time for 3 min in comparison with the Scheduler. |
| SN2026aaiv | 22:37:06 | 34:24:35 | 13.7 | SN | 1 | | 300 | 4:20:00 | 4:33:14 | Stare mode. Near for the telescope limit T = 14.01, M2 = -3.32. |
| AT2026zzz | 1:42:17 | -38:35:37 | 18.3 | | 1 | | 900 | 4:37:56 | 5:45:16 | Stare mode. T = 13.98, M2 = -3.32. |
| TW Pic | 5:34:51 | -58:01:41 | 14.2 | CV | 1 | | 120x25 | 5:47:30 | 6:55:33 | Stare mode. T = 12.8, M2 = -3.42. Change the Focus due to Temp. |
| 1eRASSJ025910.8-390331 | 2:59:11 | -39:03:31 | 15.79 | | 1.5 | | 600 | 6:57:00 | 8:31:42 | Noding. T= 12.35 , M2=-3.42. obser twice. |
| GAIA21arx | 5:38:45 | -6:17:31 | 13.7 | VS | ser twice. | | 300 | 8:33:00 | 8:44:31 | Stare mode. T= 12.22 , M2=-3.42. |
| GAIA23dfq | 5:57:45 | -13:40:34 | 14.26 | YSO | | ugriz | 420 | 8:45:29 | 9:04:40 | Imag. |
| | | | | | | | | | | |
| SKYFLATS | | | | | | | | | | |
| CALOB (operations->endofnight) | | | | | | | | 11:00:00 | | |
| | | | | | | | | | | |

| Notes Scheduler & Interface |
|---|
| - When we omit an object, it does not disappear from the queue. Maybe conection problems. In some cases we have a delay for the ob in the queue in the interface (It remained at the front of the queue all night). |
| - When refreshing the page, a "Not Found" error appears. You have to go to another section and then go back. |
| - The scheduler does not take into account the limit of the telescope and not skip the obs near. |

# SoXS daily log

| 25 Sep 2026 | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Object | RA | Dec | Mag | Spec. | slit w | Filte | Exp (s) | Start (UT | End (UT | Notes Obs |
| | | | | | | | | | | Observers: Kevin Alabarta and Camila Cárdenas |
| Open Telescope (22:54:16) :) | | | | | | | | | | |
| FLAT | | | | | | | | 22:59:16 | 23:21:00 | T=14.7, good conditions, open the telescope normaly |
| IMAGE ANALYSIS | | | | | | | | 23:22:00 | 23:57:36 | The telescope has problems take high values for M2 the operator created a ticket for technical review. He tried using the image with the most interactions on the Zenith. T=14.73, M2=-3.38 |
| LTT7987 | 20 10 56.85 | -30 13 06.6 | 12.24 | | 5 | V | 180 | 0:01:54 | 0:23:20 | T=14.73, M2=-3.38 |
| Restart the mirror, as the image analysis is not functioning correctly; the margin of error is too high, and there is a risk of the mirror breaking | | | | | | | | | | |
| ONECAL | | | | | | | | 0:38:32 | 0:42:33 | T= 14.18, M2=-3.35. |
| V4641Sgr | 0:46:26 | -25:24:26 | 13.7 | XRB | 1 | | 600 | 0:58:25 | 1:20:08 | T=13.73, M2=-3.38 |
| MAXI J1820+070 | 18:20:22 | 7:11:07 | 18.5 | XRB | 1.5 | | 1800 | 1:24:13 | 2:00:41 | T=13.65, M2=-3.36, airmass=1.7 |
| The scheduler did not load the OBs | | | | | | | | | | |
| HR Del | 8:42:20 | 19:09:39 | 12.4 | CV | | ugriz | 600 | 2:40:12 | 2:48:50 | M2=-3.39, T=13.34 |
| Atlas26ldw | 22:23:52 | -35:27:48 | 18.95 | | 1 | | 900 | 2:53:00 | 4:04:07 | M2=-3.39, T=13.34 |
| AT2026acxa | 21:02:30 | -19.27:41.76 | 15.24 | | 1 | | 300 | 4:08:16 | 4:24:48 | M2=-3.39, T=13.44 |
| AT2026aasr | 18:50:17 | -70:16:20 | 17.37 | | 1 | | 1200 | 4:56:36 | 5:44:49 | M2=-3.36, T=14.12. Change from 4 exposures of 600s to 2 exposures of 1200s each. |
| AT2022yxs | 5:39:32 | -7:22:24 | 16.6 | VS | 1 | | 600 | 5:42:03 | 6:23:44 | M2=-3.36, T=14.36. |
| AT2022yxs | 5:39:32 | -7:22:24 | 16.6 | VS | | ugriz | 600 | 6:26:45 | 6:36:50 | M2=-3.36, T=14.13. |
| SN2025pht | 4:41:29 | -2:51:56 | 20.5 | SN | 1.5 | | 2700 | 6:40:22 | 7:50:45 | M2=-3.41, T=13.03, Ang= 45° |
| SN2025pht | 4:41:29 | -2:51:56 | 20.5 | SN | 1.5 | | 2700 | 7:51:16 | 8:52:53 | M2=-3.38, T=13.54, Ang= 45° |
| | | | | | | | | | | |
| SKYFLATS | | | | | | | | | | |
| CALOB (operations->endofnight) | | | | | | | | | | |

| Notes Scheduler & Interface |
|---|
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| |
| The Scheduler didn´t send to the VOT any source for 38 minutes. It only sent one OB to the P2 but the sources didn´t have a good airmass. At the start of the night, we had 14 OBs, but during the 38 minutes of wait, we reached 52 OBs in the scheduler interface. |
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# SoXS 时域光谱仪

## SOXS GTO on extragalactic transients (excluding SNe)

This proposal groups the SOXS GTO science on extragalactic transients, not directly related to Supernovae.

180 n/y, 49% belongs to Italian group

| Proposal PI | # triggers (individual sources) | WG- | WG | TOTAL TIME AWARD (hr on source) |
|---|---|---|---|---|
| Dotto_NEO | 40 | WG- | 1 | 24,0 |
| Dotto_comets | 15 | WG- | 1 | 10,0 |
| Dotto_FRTs | | WG- | 1 | 0,9 |
| Giannini | 30 | WG- | 2 | 20,0 |
| Ripepi | 40 | WG- | 2 | 6,0 |
| Montalto | 158 | WG- | 2 | 26,0 |
| Nascimbeni | 2 | WG- | 2 | 10,0 |
| Zaggia | 2 | WG- | 2 | 0,0 |
| Tsygankov | 1 | WG- | 3 | 3,6 |
| Baglio_NS_early | 5 | WG- | 3 | 0,5 |
| Baglio_NS_rise | 10 | WG- | 3 | 3,1 |
| Baglio_NS_peak | 1 | WG- | 3 | 4,0 |
| Baglio_NS_decay | 5 | WG- | 3 | 1,8 |
| Baglio_NS_reflare | 2 | WG- | 3 | 0,2 |
| Baglio_BH_outburst | 5 | WG- | 3 | 0,5 |
| Baglio_BH_early | 7 | WG- | 3 | 0,2 |
| Baglio_BH_UL | 40 | WG- | 3 | 15,0 |
| Campana_magnetar | 1 | WG- | 3 | 0,3 |
| deMartino | 4 | WG- | 4 | 7,0 |
| Neustroev | 11 | WG- | 4 | 5,0 |
| Izzo | 10 | WG- | 4 | 24,0 |
| Ben-Ami | 4 | WG- | 4 | 10,0 |
| Stritzinger | 9 | WG- | 5 | 7,5 |
| Stritzinger | 8 | WG- | 5 | 3,0 |
| Stritzinger | 10 | WG- | 5 | 10,0 |
| Stritzinger | 2 | WG- | 5 | 4,0 |
| Stritzinger | 3 | WG- | 5 | 1,9 |
| Badash | 60 | WG- | 5 | 20,0 |
| Chen | 8 | WG- | 5 | 5,0 |
| Arcavi_14hls | 1 | WG- | 6 | 5,0 |
| FBOT (Campana, Nicholl, Bauer) | 4 | WG- | 6 | 74,3 |
| Nicholl_SLSN | 3 | WG- | 6 | 22,0 |
| Kangas_LSN | 2 | WG- | 6 | 5,0 |
| Kotak | 6 | WG- | 7 | 43,0 |
| Kankare | 1 | WG- | 8 | 5,0 |
| Kangas_LSN | 3 | WG- | 8 | 8,0 |
| GalYam_Exotic | 3 | WG- | 8 | 8,0 |
| GalYam_Exotic | 3 | WG- | 8 | 1,0 |
| GalYam_Exotic | 3 | WG- | 8 | 2,0 |
| GalYam_Nearby | 30 | WG- | 8 | 178,0 |
| Campana_SBO | 40 | WG- | 8 | 20,0 |
| Tartaglia | 12 | WG- | 8 | 69,0 |
| Campana_IcBL | 2 | WG- | 8 | 30,0 |
| Marziani | 1 | WG- | 9 | 0,0 |
| Rigamonti | 2 | WG- | 9 | 0,0 |
| Sbarrato | 4 | WG- | 9 | 16,8 |
| Arcavi_TDE | 100 | WG- | 10 | 28,9 |
| Arcavi_TDE | 10 | WG- | 10 | 36,3 |
| Mattila | 4 | WG- | 10 | 10,0 |
| Campana_FRB | 1 | WG- | 11 | 1,0 |
| GRB-FXT (D'Avanzo, Campana) | 100 | WG- | 11 | 108,0 |
| Campana_GWspe | 10 | WG- | 12 | 5,0 |
| Campana_GWpho | 10 | WG- | 12 | 18,0 |
| Campana_neutrino | 10 | WG- | 12 | 2,0 |

GRBs, firmly establishing their connection with the core collapse of massive stars. We aim to observe any GRB visible from La Silla whose position is known with a precision of less than $\sim 2'$ (radius, 90% c.l.) to enable a search for/detection of the optical afterglow with the acquisition camera. Sources of the initial trigger can be any high-energy satellite (e.g. *Swift, SVOM Fermi, Einstein Probe, HERMES*) or GRB afterglows found independently, even without a high-energy trigger (i.e., 'orphan' afterglows). Systematic optical spectroscopic follow-up of GRBs

• **GW multimessenger astronomy.** The single case of GW170817 demonstrates the vital importance of time-resolved optical/NIR imaging and spectroscopic follow-up of electromagnetic candidate counterparts of GW triggers. The scientific return will be enormous. New events will provide a stress test for kilonova emission models. Even if rarer (but detectable to a larger distance due to the larger mass involved), BH-NS merger events will further strain kilonova models. We plan to follow spectroscopically (if $r \lesssim 21$) and photometrically (if $r \lesssim 24$) all KNe serendipitously discovered not connected to short GRBs. According to Andreoni et al. (2022), we might expect $\sim 1$ kilonova/yr within 100 Mpc detected by Rubin. At $r \sim 23$ we can sample kilonova out to $\sim 500$ Mpc and we might expect 2-20 kilonovae/yr with Rubin.

| Object | Trigger. mag | Constraints | Time* |
|---|---|---|---|
| BL Lac | – | TeV flare | |
| FSRQ | – | Flux 0.1-100 GeV, $> 1 \times 10^{-7}$ ph cm$^{-2}$ s$^{-1}$ | **17hr** |
| ANT | $M < -20$ | or $t_{\rm rise} > 20$ d (nuclear) | **10hr** |
| TDE Discovery | $r < 20$ | blue, rising, nuclear | |
| TDE follow-up | | confirmed TDE | **67hr** |
| Long GRB | $r < 21$ or $H < 19$ | High energy trigger or afterglow - spectrum | |
| Long GRB | $r > 21$ | High energy trigger or afterglow -photometry | |
| GRB-SN | $r < 21$ or $H < 19$ | High energy trigger or afterglow $z \lesssim 0.2$ | |
| Short GRB | $r < 21$ or $H < 19$ | High energy trigger or afterglow - spectrum | |
| Short GRB | $r > 21$ | High energy trigger or afterglow - photometry | **138hr** |
| (FXT) | $r < 21$ or $H < 19$ | High energy trigger or afterglow - spectrum | |
| (FXT) | $r > 21$ | High energy trigger or afterglow - photometry | |
| GW/Kilonova spectrum | $r < 21$ | GW/Kilonova detection not connected to SGRB | |
| GW/Kilonova photometry | $r < 24$ | GW/Kilonova detection not connected to SGRB | |
| HEN Neutrino | $r < 21$ | Gold HEN & counterpart | **26hr** |

# SoXS 时域光谱仪

- SoXS 的所有观测都是 ToO，通过 SoXS Marshall 系统提交 https://drive.google.com/file/d/1l_Fs0ZqnkbQiUT6Y0hfRM_VGdofT3Sje/view?usp=drive_web

- Soft/Hard ToO：自提交后 7/3 天内完成观测（与 ESO 的协定）

- Urgent OB：打断当前的观测，立即执行该观测（原则上）
  至今仍未有能够满足条件的源来测试该模式

# SoXS 时域光谱仪

- SoXS
  https:
  e/view
- Soft/
- Urgen
  至今

How should it work.
- There is a GRB.
- People from the satellite, when they are *sure* it is a GRB, and check the visibility from La Silla contact Kevin/Cristina/Giulia (whatsapp, phone, email?; observers please share your preferred system)
- People from the satellite shares via email coordinates (so not to type them wrongly) and a finding chart, possibly with the error circle
- Kevin/Cristina/Giulia use the Scheduler interface to generate an Urgent OB
- As standard parameters I would go with
  Acquisition r 120s (SNR=4 for r=21)
  Spectrum Stare 2 observations each UVVIS:900s; NIR: DIT=900s NDIT=1 / slit 1" or 1.5" (better suggestions welcome, this is not to saturate in case of bright r~15 GRB) -- NIR is not working nowadays (ESO fault), hope to recover late in the run
- As soon as the current observation ends, the urgent OB should start
- Comparing the finding chart with the SOXS image, one should spot the intruder ("trova l'intruso") within the error circle (or slightly outside)
- If there is such an object go for it in spectroscopy, otherwise abort the observation

dofT3Sj

协定）

# Urgent OB 的例子

## [EP Alert] 2026-09-26T11:52:25Z + Complete

H

**EP Alert** <zhou1997810@gmail.com>

9月26日周六 13:59 (2天前)

发送至 zhou1997810， wangyun_95， kevin.alabartajativa， raya.dastidar， maria.baglio， giulia.illiano

WXT Trigger Time: 2026-09-26T11:52:25 UTC
WXT R.A., Dec. (J2000) = 6.447, -9.954 +/- 3'

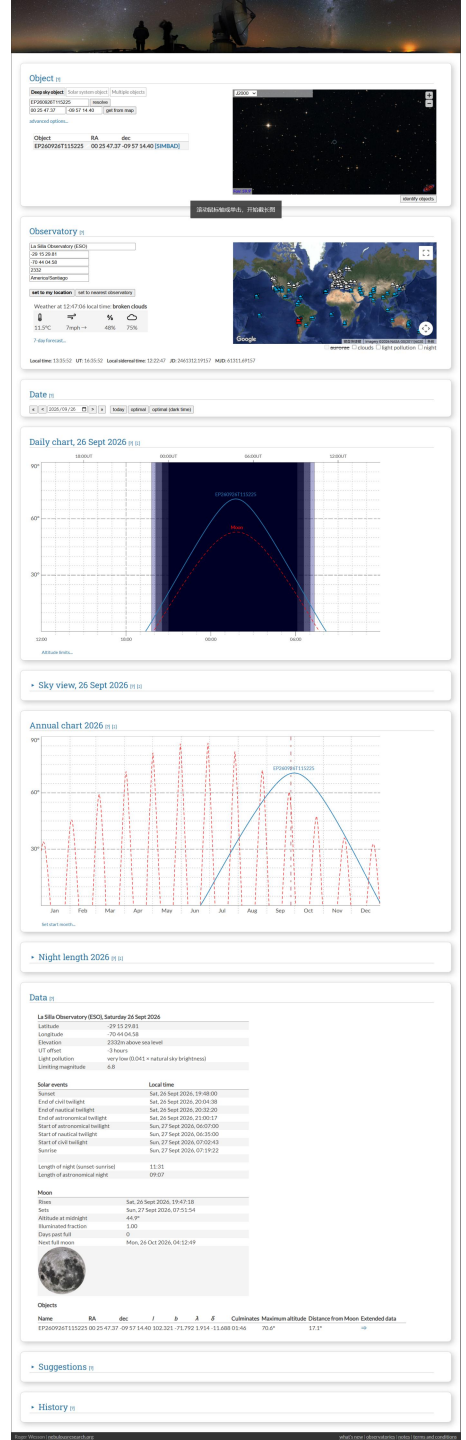
FXTA Tigger Time: 2026-09-26T11:56:51 UTC
FXTA R.A., Dec. (J2000) = 6.4628, -9.9655 +/- 20"
Distance to the WXT position = 70.4"

FXTB Tigger Time: 2026-09-26T11:56:51 UTC
FXTB R.A., Dec. (J2000) = 6.4633, -9.9654 +/- 20"
Distance to the WXT position = 71.6"
==========

Visibility
La Silla: https://airmass.org/chart/obsid:lasilla/date:2026-09-26/object:EP260926T115225/ra:6.447/dec:-9.954
Lick: https://airmass.org/chart/obsid:lick/date:2026-09-26/object:EP260926T115225/ra:6.447/dec:-9.954
Keck: https://airmass.org/chart/obsid:keck1/date:2026-09-26/object:EP260926T115225/ra:6.447/dec:-9.954

8'x8' DSS2-red image centered on the WXT position (copy the link to your browser if it does not download the .fits file automatically):
http://archive.eso.org/dss/dss/image?ra=6.447&dec=-9.954&equinox=J2000&name=&x=8&y=8&Sky-Survey=DSS2-red&mime-type=download-fits&statsmode=WEBFORM

Content of the ds9 region file:
# Region file format: DS9 version 4.1
global color=green dashlist=8 3 width=1 font="helvetica 10 normal roman" select=1 highlite=1 dash=0 fixed=0 edit=1 move=1 delete=1 include=1 source=1
fk5
circle(6.447,-9.954,180.000") # text={WXT - 3' (90% radius)} dash=1
circle(6.4628,-9.9655,20.000") # text={FXTA - 20" (90% radius)}
circle(6.4633,-9.9654,20.000") # width=3 font="helvetica 10 bold roman" text={FXTB - 20" (90% radius)}



# Urgent OB 的例子

## [EP Alert] 2026-09-26T11:52:25Z + Complete

H **EP Alert** <zhou1997810@gmail.com>
发送至 zhou1997810， wangyun_95， kevin.alabartajativa， raya.dastidar， maria.baglio， giulia.illiano

WXT Trigger Time: 2026-09-26T11:52:25 UTC
WXT R.A., Dec. (J2000) = 6.447, -9.954 +/- 3'

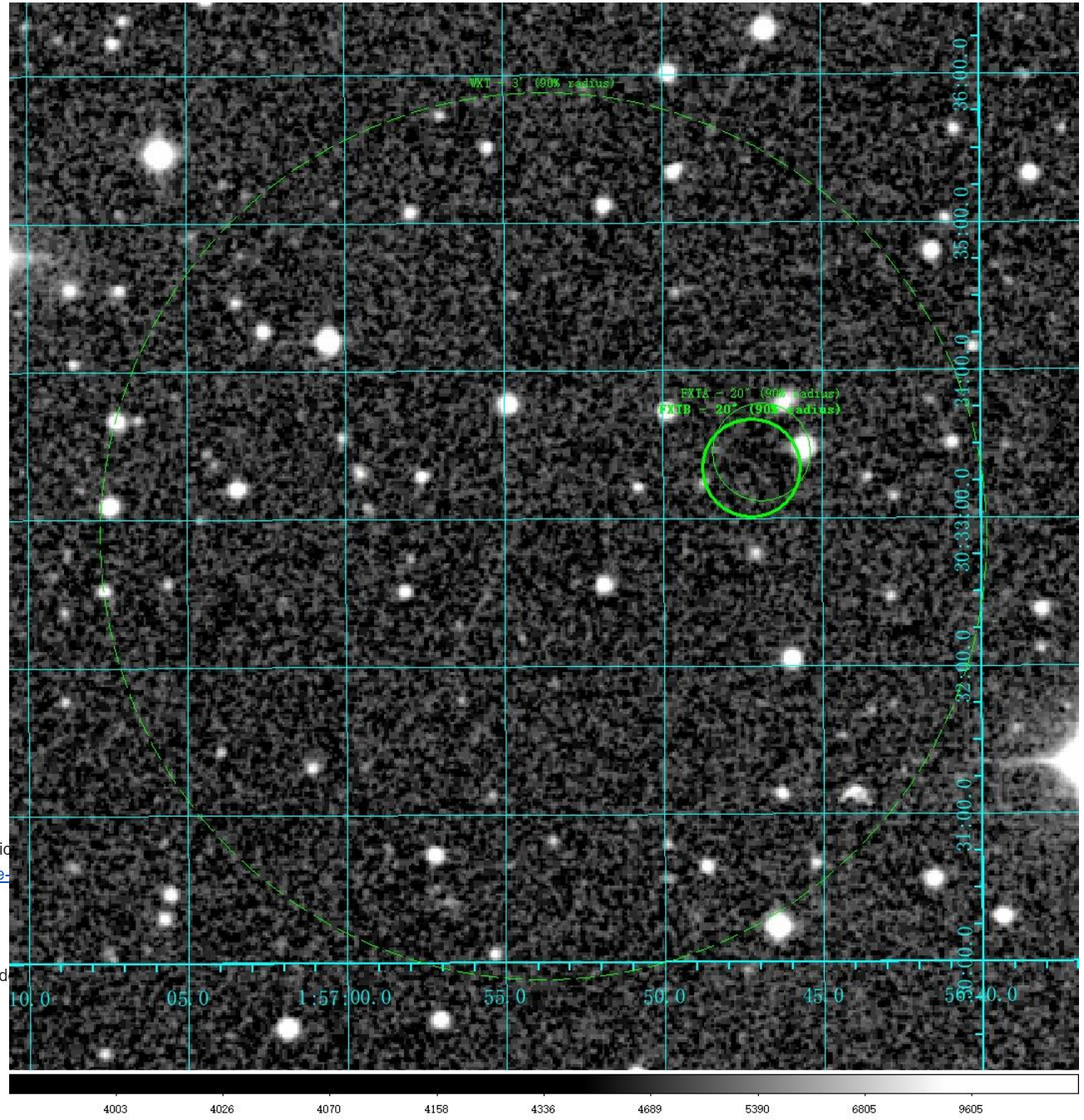
FXTA Tigger Time: 2026-09-26T11:56:51 UTC
FXTA R.A., Dec. (J2000) = 6.4628, -9.9655 +/- 20"
Distance to the WXT position = 70.4"

FXTB Tigger Time: 2026-09-26T11:56:51 UTC
FXTB R.A., Dec. (J2000) = 6.4633, -9.9654 +/- 20"
Distance to the WXT position = 71.6"
==========

Visibility
La Silla: https://airmass.org/chart/obsid:lasilla/date:2026-09-26/object:EP260926T115225/ra:6.447/dec:-9.954
Lick: https://airmass.org/chart/obsid:lick/date:2026-09-26/object:EP260926T115225/ra:6.447/dec:-9.954
Keck: https://airmass.org/chart/obsid:keck1/date:2026-09-26/object:EP260926T115225/ra:6.447/dec:-9.954

8'x8' DSS2-red image centered on the WXT position (copy the link to your browser if it does not download the .fits file automatic
http://archive.eso.org/dss/dss/image?ra=6.447&dec=-9.954&equinox=J2000&name=&x=8&y=8&Sky-Survey=DSS2-red&mime-

Content of the ds9 region file:
# Region file format: DS9 version 4.1
global color=green dashlist=8 3 width=1 font="helvetica 10 normal roman" select=1 highlite=1 dash=0 fixed=0 edit=1 move=1 d
fk5
circle(6.447,-9.954,180.000") # text={WXT - 3' (90% radius)} dash=1
circle(6.4628,-9.9655,20.000") # text={FXTA - 20" (90% radius)}
circle(6.4633,-9.9654,20.000") # width=3 font="helvetica 10 bold roman" text={FXTB - 20" (90% radius)}



# LSST 时代下的挑战

VERA C. RUBIN OBSERVATORY

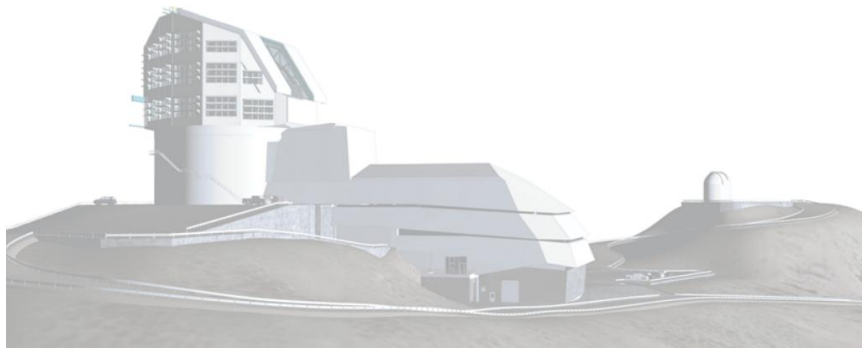
Vera C. Rubin Observatory
Data Management

**LSST Alerts: Key Numbers**

Melissa L. Graham, Eric C. Bellm, Leanne P. Guy, Colin T. Slater,
Gregory P. Dubois-Felsmann, Mario Jurić, and
Rubin Observatory Data Management System Science Team

DMTN-102

https://doi.org/10.71929/rubin/2997858
Latest Revision: 2024-04-24



- Supernovae: LSST is predicted to observe a total of ~10 million supernovae in 10 years, or ~1 million per year. Since SNe are typically only visible for a few months, there might be ~0.3 million detectable at any given time. Over 15,000 $\mathrm{deg}^2$ of extragalactic survey area, that's ~20 SNe $\mathrm{deg}^{-2}$ or ~200 alerts for SNe per visit.

② **科学目标与总体事件量推导来源：**

- **文献名称：** LSST Science Book, Version 2.0 (arXiv:0912.0201) Wikipedia
- **主要章节：** Chapter 8 (*Exploring the Transient Optical Sky*) 与 Chapter 11 (*Supernovae*)
- **核心贡献：** 白皮书中系统估算了 LSST 10 年巡天期间的各类爆发事件总数： Wikipedia
  - **超新星总数：** 约 **1000 万个**（$10^7$），即平均每年发现约 **100 万个** 新超新星。
  - 平均到全天区 $15,000\ \mathrm{deg}^2$ 的河外巡天面积，每年每平方度约产生 60～70 个新超新星；折算到单晚（考虑观测覆盖率与重访周期），即 **~1–3 个新爆发河外暂现源 / $\mathrm{deg}^2$ / night**。

考虑到 GW 定位几十 – 几百平方度的定位区间
暂现源的数量实在太多

# SoXS 光谱后随的策略

**一些客观事实**

- LSST 扫描 GW 天区时会有巨量的 EM 对应体候选
- EM 对应体是类似 KN 的，颜色由蓝转红，总体光度 ~42
- EM 对应体的宿主星系应该是年老的星系
- 外激波可能会掩盖类似 KN 的辐射

**观测策略**

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**观测策略**

- 利用星表快速锁定一些在年老星系附近发现的暂现源，它们更有可能是 EM 对应体
- 如果在第一批次的光谱数据中没有找到 EM 对应体，重点关注存在颜色转换的源
- 总体上从亮至暗源排查，节约观测时间

# SoXS 光谱后随的策略

- 年老星系筛选规则：
  - 星系距离是否与 GW 光度距离匹配
  - 假设该暂现源的宿主就是我们认为的宿主，其光度是否 $< \sim 10^{42}$ erg/s
  - 如果有接近同期的测光结果，该源的颜色是否是类 KN 的
- 颜色转换筛选规则：
  - 在 1 – 3 天内，源的颜色存在着显著的蓝 –> 红
  - 附近是否存在与 GW 光度距离匹配的星系
  - 光度匹配
- 亮暗源观测顺序：
  - 亮的源可以用更短的时间得到靠谱的光谱，但也要考虑可见性来具体编排

# 总结

- 当前已经建成了许多大视场巡天设备，暂现源的发现效率已经大大提高
  - 在成像中发现 GW 电磁对应体候选体能力有了长足的进步
  - 强大的巡天能力将带来许多污染源，尤其是对于距离较远的 GW 事件
- 光谱后随能力严重不足，证认能力匮乏
  - 专门为暂现源设计的光谱仪 SoXS，其观测速度也仅为 >~10 每晚
  - 更昂贵先进的设备观测时间更紧张
- 通过分析 EM 对应体的物理性质，我们或许可以排除掉大量的暂现源，将真正候选体的数量控制在 SoXS 的承受范围？